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GARBAGE WASTE CLASSIFICATION USING CNN AND TRANSFER LEARNING (MOBILENETV2)

1. Introduction

Waste management is a critical challenge in modern society, especially with the increasing volume of waste generated daily. Efficient waste segregation is essential for recycling, reducing environmental pollution, and promoting sustainability. Manual segregation is time-consuming, error-prone, and inefficient.

This project presents an automated garbage classification system using deep learning techniques. The system leverages a convolutional neural network (CNN) architecture, specifically MobileNetV2, to classify waste images into predefined categories. The goal is to build an efficient, accurate, and scalable solution for real-world waste management applications.

2. Problem Definition

To design and implement an image classification system capable of categorizing garbage images into multiple classes such as plastic, paper, metal, etc.

3. Motivation

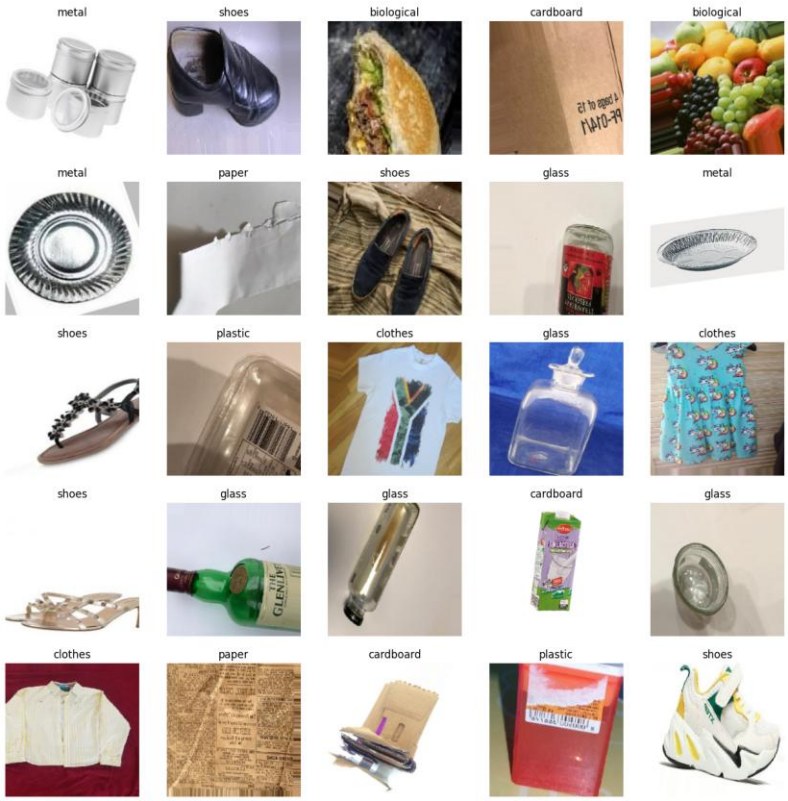
- Improve recycling efficiency
- Reduce human effort in waste sorting
- Enable smart waste management systems
- Support environmental sustainability initiatives

Real-World Applications:

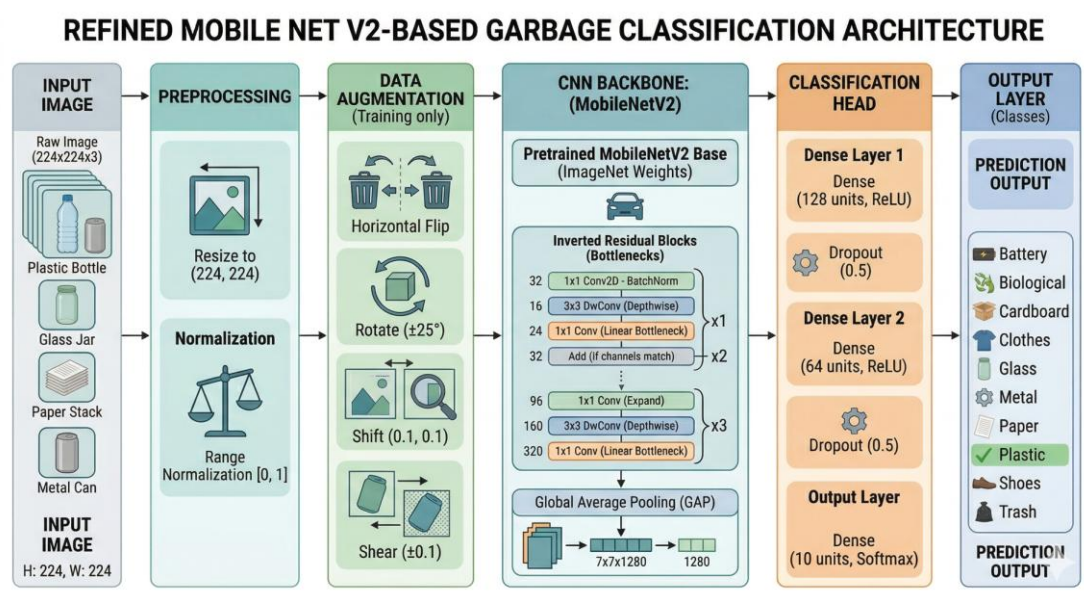
- Smart bins with automated classification
- Recycling plants
- Mobile applications for waste identification
- Smart city infrastructure

4. Dataset

The dataset contains ~13,000 images across 10 classes including battery, biological, cardboard, clothes, glass, metal, paper, plastic, shoes, and trash.



5. System Architecture



- Input Image (224×224×3)
- Preprocessing (Resizing, Normalization)
- Data Augmentation (Training only)
- Feature Extraction using MobileNetV2
- Classification using Dense Layers
- Output Prediction (10 Classes)

6. Model Design

Base Model: MobileNetV2

- Pretrained on ImageNet
- Lightweight and efficient
- Uses depth wise separable convolutions

Model Architecture:

- Frozen base layers initially
- Global Average Pooling layer
- Fully connected layers:
    - Dense (128 units, ReLU)
    - Dropout (0.5)
    - Dense (64 units, ReLU)
    - Dropout (0.5)
- Output layer:
    - Dense (10 units, Softmax)

Fine-Tuning:

- Last 50 layers of MobileNetV2 unfrozen
- Learning rate reduced to improve performance

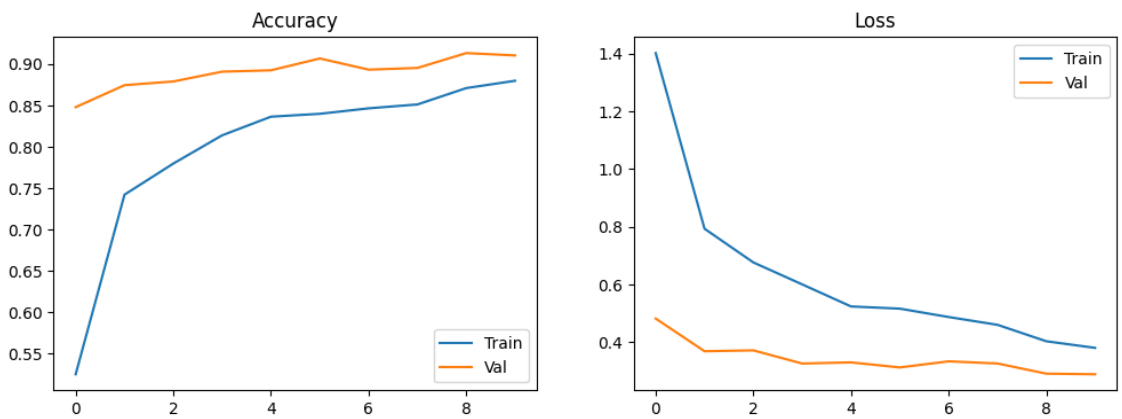
7. Training Configuration

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Metrics:
    - Accuracy
    - Top-3 Accuracy
- Batch Size: 32
- Epochs: 10 + 5 (fine-tuning)

Callbacks Used:

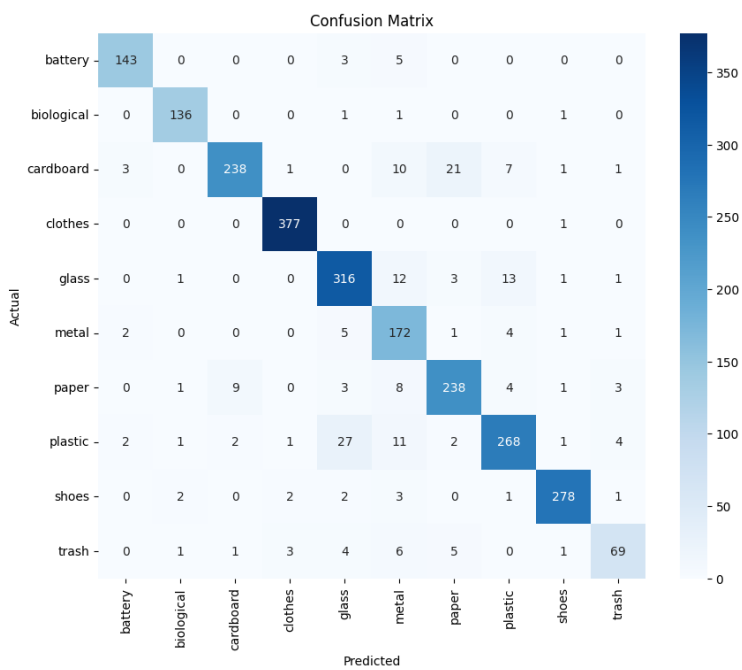
- Early Stopping
- Model Checkpoint
- Reduce Learning Rate on Plateau

8. Experimental Results



- Final Training Accuracy: ~87%
- Final Validation Accuracy: ~91%
- Validation Loss: ~0.29

9. Confusion Matrix Analysis



10. Challenges Faced

- Class imbalance in dataset
- Similar visual features between classes
- Limited dataset size
- Overfitting risk
- Variability in lighting and angles

11. Conclusion

This project successfully demonstrates the development of an end-to-end garbage classification system using deep learning. By leveraging MobileNetV2 and transfer learning, the model achieves high accuracy while remaining computationally efficient.

The system can be extended for real-world deployment in smart waste management systems, contributing to environmental sustainability and automation.

12. Future Work

- Use larger dataset
- Apply advanced architectures (EfficientNet)
- Deploy model using Flask or FastAPI
- Real-time classification using camera